Lab 2.2: Custom Reusable Package (String and Math Functions)

utils/utils.go

```go
package utils

import "strings"

func Reverse(s string) string {
    runes := []rune(s)
    for i, j := 0, len(runes)-1; i < j; i, j = i+1, j-1 {
        runes[i], runes[j] = runes[j], runes[i]
    }
    return string(runes)
}

func CountVowels(s string) int {
    count := 0
    for _, ch := range strings.ToLower(s) {
        if strings.ContainsRune("aeiou", ch) {
            count++
        }
    }
    return count
}

func Factorial(n int) int {
    if n < 0 {
        return -1
    }
    result := 1
    for i := 2; i <= n; i++ {
        result *= i
    }
    return result
}

func Power(base, exp int) int {
    result := 1
    for i := 0; i < exp; i++ {
        result *= base
    }
    return result
}
```

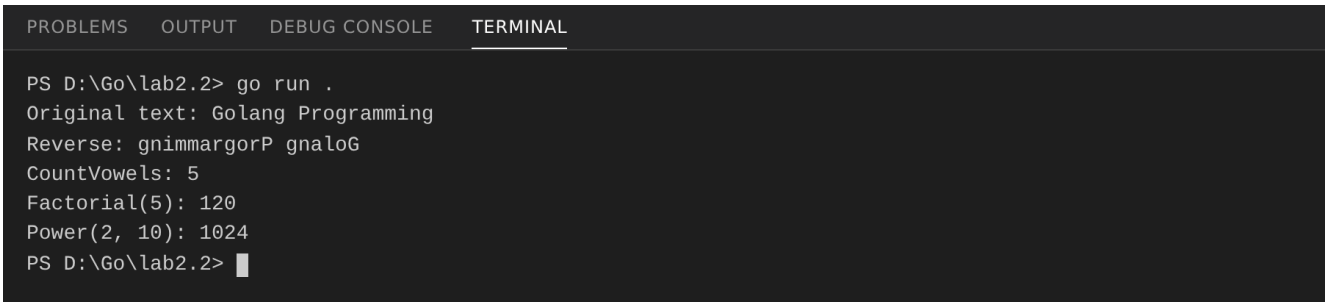
main.go

```go
package main

import (
    "fmt"

    "lab2.2/utils"
)

func main() {
    text := "Golang Programming"
    fmt.Println("Original text:", text)
    fmt.Println("Reverse:", utils.Reverse(text))
    fmt.Println("CountVowels:", utils.CountVowels(text))

    fmt.Println("Factorial(5):", utils.Factorial(5))
    fmt.Println("Power(2, 10):", utils.Power(2, 10))
}
```

Lab 2.2: Output